A Machine Learning Approach to Predicting No-Shows in a Philippine Pediatric Therapy Clinic

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ABSTRACT

This study develops a binary classification model to predict patient no-shows in a Philippine pediatric therapy clinic by analyzing operational clinic data combined with external environmental factors. The research addresses the critical challenge of appointment non-attendance, which leads to significant revenue loss and inefficient therapist utilization. By examining appointment schedules, payment histories, patient demographics, local weather events, and Philippine holidays, the study aims to identify the strongest predictors of non-attendance and uncover specific high-risk appointment profiles. The dataset comprises 1,506 appointments with an overall no-show rate of 42.4%. Three machine learning algorithms were implemented and compared: logistic regression (AUC = 0.877), random forest (AUC = 0.892), and gradient boosting (AUC = 0.919). SHAP analysis revealed that typhoon signal level, historical attendance behavior, and lead time were the most influential predictors. Hypothesis testing confirmed that environmental factors and temporal patterns significantly affect attendance, while surprisingly, financial commitment through down payments showed no significant effect. These findings provide actionable insights for clinic management to reduce no-show rates through targeted intervention strategies focused on weather-related flexibility, strategic scheduling, and patient history-based risk assessment.

Index Terms*—Appointment no-shows, binary classification, machine learning, pediatric therapy, predictive modeling, SHAP analysis, healthcare analytics, Philippines*

I. INTRODUCTION

Appointment of non-attendance represents a persistent challenge in healthcare delivery systems worldwide, with significant implications for operational efficiency, revenue sustainability, and patient care quality. In pediatric therapy settings, where consistent attendance is crucial for developmental progress and treatment effectiveness, the problem of no-shows becomes particularly acute. The phenomenon of missed appointments generates cascading effects throughout the healthcare ecosystem: therapists experience fragmented schedules with unused time slots that could have been allocated to other patients, clinics face revenue losses from unpaid services that cannot be recovered, and patients themselves miss critical intervention opportunities during sensitive developmental windows. While this challenge exists across healthcare systems globally, it manifests with unique characteristics in the Philippine context, where factors such as tropical weather disruptions, cultural holiday patterns, and economic constraints create a distinct attendance behavior profile that remains underexplored in existing literature.

This study addresses the critical gap in understanding appointment adherence behavior specific to Philippine pediatric therapy clinics by developing a machine learning-based prediction model for no-shows. The research transforms routine administrative data, typically collected but underutilized, into actionable intelligence that can guide clinic management decisions. By systematically analyzing appointment schedules, payment behaviors, patient demographics, and external environmental factors, the study seeks to identify which variables most strongly predict non-attendance and how these variables interact to create high-risk appointment scenarios. The investigation is grounded in five primary hypotheses examining the roles of financial commitment, temporal scheduling patterns, environmental disruptions, demographic risk profiles, and historical cancellation behavior in determining attendance outcomes.

The significance of this research extends beyond academic contribution to practical clinical applications. Unlike previous studies conducted primarily in Western healthcare systems, this investigation incorporates culturally and geographically specific variables relevant to the Philippine context, including typhoon signal warnings, local government-declared holidays, and payment structures common in private therapy practices. The model employs advanced interpretability techniques through SHAP analysis to quantify and rank predictor importance, enabling clinic administrators to understand not merely which factors matter, but how much they matter and in what direction they influence attendance decisions. Furthermore, by examining the interaction effects among financial, environmental, and demographic variables, the research aims to identify specific high-risk appointment profiles that warrant targeted intervention strategies.

II. LITERATURE REVIEW

The application of machine learning techniques to predict healthcare appointment no-shows has gained considerable attention in recent years, driven by the dual objectives of improving operational efficiency and enhancing patient care delivery. This literature review examines four seminal studies that establish the theoretical foundation and methodological precedents for the current research, while simultaneously highlighting critical gaps that this investigation addresses within the Philippine pediatric therapy context.

A. Demographic and Behavioral Predictors in Pediatric Settings

Dunstan et al. [1] conducted a comprehensive analysis of nearly 396,000 pediatric appointments in a Chilean hospital setting, demonstrating that demographic variables, socioeconomic status, and prior attendance behavior serve as strong predictors of missed visits. The study employed gradient boosting machines and random forest algorithms to achieve prediction accuracies exceeding 75%, with historical cancellation patterns emerging as the most influential predictor. Their findings established that patients with previous cancellations exhibited significantly higher probabilities of subsequent no-shows, a behavioral consistency that suggests attendance patterns are relatively stable over time. Additionally, the research identified distance from the clinic and wait times as important factors, with longer distances and extended waiting periods both correlating with increased cancellation rates.

The Chilean study provides valuable methodological guidance for the current research, particularly in its emphasis on historical cancellation data and demographic characteristics as core predictive features. However, several critical limitations restrict its direct applicability to the Philippine context. First, the study did not incorporate environmental variables such as weather disruptions or seasonal patterns, factors that may be particularly relevant in tropical climates characterized by typhoon activity. Second, the research examined a public hospital setting where payment mechanisms differ substantially from private therapy clinics. The absence of financial commitment variables, such as down payment status or payment plan adherence, represents a significant gap given that private healthcare delivery often operates under different economic incentives than public systems. Finally, cultural factors specific to the Philippines, including local holiday patterns and family decision-making structures around pediatric care, remain unexplored.

B. Explainable AI and Temporal Analysis

Hathaway et al. [2] advanced the field by applying explainable AI techniques to pediatric appointment data spanning pre-pandemic and pandemic periods. Their use of XGBoost models combined with SHAP values for feature interpretation represents a methodological innovation that addresses the "black box" criticism often leveled at machine learning approaches in healthcare. The study revealed that appointment characteristics, including time of day and day of week, exhibited strong predictive power, with early morning and late afternoon time slots showing higher no-show rates. Furthermore, the temporal analysis across different pandemic phases demonstrated that prediction models require periodic recalibration to maintain accuracy as healthcare delivery contexts evolve.

This research provides crucial validation for employing SHAP analysis in the current study, as it demonstrates both the technical feasibility and clinical interpretability of this approach in pediatric settings. The emphasis on temporal patterns aligns with the current research's hypothesis regarding time-based scheduling effects. However, the study's geographic limitation to a non-Philippine setting means it does not capture culturally specific temporal patterns, such as increased cancellations during Filipino holiday seasons or typhoon seasons. Additionally, while the pandemic analysis offers insights into contextual disruptions, it does not systematically examine how routine environmental factors, as opposed to extraordinary events, influence attendance behavior under normal operating conditions.

C. Environmental Variables and Deep Learning Approaches

Liu et al. [3] introduced environmental factors into no-show prediction by incorporating local weather data into their models, demonstrating that adverse weather conditions significantly influenced appointment attendance in a U.S. pediatric setting. Their deep learning architecture, benchmarked against logistic regression baselines, achieved strong performance even with missing data through the application of multiple imputation techniques. The inclusion of precipitation levels, temperature extremes, and severe weather alerts as predictor variables represented an important expansion of the feature space beyond traditional demographic and behavioral variables. The study found that severe weather events could increase no-show probabilities by up to 40% compared to normal weather conditions.

This research establishes precedent for incorporating environmental variables into prediction models, a key component of the current study's approach. However, the specific weather patterns examined in the U.S. context differ substantially from Philippine tropical climate conditions. The current research extends this work by incorporating typhoon signal warnings and rainy season indicators, environmental factors with particular salience in the Philippine archipelago. Additionally, the study did not examine government-declared holidays or cultural festival periods, which may exhibit different attendance dynamics than weather-related disruptions alone. The current research addresses this gap by systematically coding Philippine national and local holidays as distinct predictor variables.

D. Historical Behavior and Gradient Boosting Models

Mochón et al. [4] focused specifically on the predictive power of historical attendance patterns using gradient boosting algorithms. Their research demonstrated that incorporating a patient's past appointment history significantly improved model accuracy compared to models relying solely on current appointment characteristics. The study found that the number of previous cancellations, the recency of the last cancellation, and the consistency of attendance patterns all contributed independently to prediction performance. Models including these historical features achieved AUC scores above 0.80, compared to 0.65 for models without historical data.

This emphasis on historical behavior informs the current study's hypothesis regarding the influence of prior cancellation patterns on future attendance. However, the study's relatively limited feature set, excluding both environmental factors and financial variables, represents a significant constraint. The research did not examine whether financial commitment mechanisms, such as required deposits, might moderate the relationship between historical behavior and future attendance. Additionally, the study lacked consideration of external contextual factors that might explain why historically reliable patients might deviate from established patterns, such as holiday periods or weather emergencies.

E. Research Gap and Current Study Contribution

While these studies collectively establish robust methodological frameworks for no-show prediction, they share common limitations that the current research addresses. First, none were conducted in Philippine healthcare settings, leaving unexamined how culturally and geographically specific factors influence attendance behavior in this context. Second, financial commitment variables, particularly relevant in private therapy practices where payment structures differ from public or insurance-based systems, remain largely unexplored in existing literature. Third, the interaction between environmental, financial, and demographic factors has not been systematically examined to identify high-risk appointment profiles that emerge from variable combinations rather than individual predictors. The current study fills these gaps by developing a prediction model specifically tailored to Philippine pediatric therapy clinics, incorporating a comprehensive feature set spanning financial, environmental, demographic, and behavioral dimensions, and employing SHAP analysis to quantify both individual feature importance and interaction effects among predictors.

III. METHODOLOGY

This section delineates the research design, data collection procedures, variable operationalizations, and analytical techniques employed to develop the no-show prediction model. The methodology emphasizes replicability and transparency to enable validation and extension by future researchers.

A. Study Design and Data Collection

This study employs a retrospective observational design, analyzing 1,506 appointments from a private pediatric therapy clinic in the Philippines. The dataset spans multiple months and captures appointments across diverse seasonal contexts including both typhoon-affected periods and normal weather conditions. The collection period encompasses several Philippine national holidays and local festival seasons, providing variation in holiday-related predictor variables. Data were extracted from the clinic's therapy schedule and client information sheets, supplemented with environmental data from official Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA) bulletins and government holiday declarations.

B. Variable Definitions

The study operationalizes variables across five conceptual categories. The *outcome variable* is appointment outcome, coded as 0 (attended) or 1 (no-show). *Financial variables* include down payment status (binary), down payment amount, and payment account status (Arrears vs. Updated). *Temporal variables* capture day of week, appointment time block, and weekend status. *Environmental variables* include holiday indicator, typhoon signal level (0-5), and rainy season indicator. *Demographic and behavioral variables* encompass age group, diagnosis category, distance from clinic, previous appointment count, previous no-show count, no-show ratio, and lead time days.

C. Data Preprocessing

The dataset underwent systematic preprocessing including handling missing values through appropriate imputation strategies, encoding categorical variables, and log-transforming highly skewed continuous variables (distance, duration, lead time, previous appointment count). The dataset was partitioned into training (70%), validation (15%), and test (15%) sets using stratified random sampling to ensure balanced outcome distribution across subsets.

D. Statistical Analysis and Modeling

The analytical strategy employed multiple complementary techniques. Descriptive statistics characterized the dataset's distributional properties. Hypothesis tests (chi-square tests for categorical variables, t-tests and ANOVA for continuous variables) assessed the statistical significance of associations between predictor variables and attendance outcomes. Three machine learning algorithms were implemented: logistic regression (baseline interpretable model), random forest (capturing non-linear relationships), and gradient boosting (potentially superior predictive performance). Model hyperparameters were optimized through cross-validation. Performance was evaluated using AUC-ROC, precision, recall, F1-score, and calibration metrics. SHAP analysis quantified feature importance and revealed interaction effects.

E. Ethical Considerations

The study adhered to ethical research principles by excluding personally identifiable information. Patient names, exact birthdates, and precise addresses were not collected. All data were stored securely with restricted access. The analysis used fully anonymized administrative data for quality improvement purposes.

IV. RESULTS

A. Dataset Characteristics

The final dataset comprised 1,506 appointments, of which 638 (42.4%) resulted in no-shows and 868 (57.6%) were attended. This substantial no-show rate underscores the severity of the attendance challenge facing the clinic. The patient population spanned ages 2 to 18 years, with the largest cohort in the 4-6 years age group (38.2%), followed by 7-9 years (28.6%). Diagnosis categories included Autism Spectrum Disorder (ASD, 31.5%), Speech and Language Delay (SLD, 24.8%), Occupational Therapy needs (18.3%), and other developmental conditions. Geographic distance from the clinic ranged from less than 1 km to over 40 km, with a median distance of 10.2 km. Approximately 67.3% of appointments had down payments recorded, with amounts ranging from ₱150 to ₱1,500.

B. Hypothesis Testing Results

1) H1: Financial Commitment

Contrary to expectations, no significant difference in no-show rates was observed between appointments with down payments (42.7% no-show rate) and those without (41.8% no-show rate). Chi-square test yielded $\chi^2 = 0.12$, $p = 0.73$, indicating that the current deposit system does not significantly incentivize attendance. Similarly, payment status (Arrears vs. Updated) showed no significant association with attendance outcomes ($\chi^2 = 1.45$, $p = 0.23$). This unexpected finding challenges conventional economic incentive assumptions and suggests that deposit amounts may be too low to create meaningful behavioral change, or that other factors override financial considerations in attendance decisions.

2) H2: Temporal Patterns

No-show rates differed significantly by day of the week ($\chi^2 = 18.42$, $p = 0.005$). Wednesdays exhibited the highest no-show rate at 52.1%, while Saturdays showed the lowest at 31.2%. Weekend appointments (Saturday-Sunday) demonstrated significantly lower no-show rates (33.4%) compared to weekdays (45.8%), $t = 4.23$, $p < 0.001$. Appointment time also influenced attendance: morning sessions (8:00-12:00) had a 46.7% no-show rate compared to 38.2% for afternoon sessions (13:00-17:00), $\chi^2 = 8.91$, $p = 0.003$. These patterns suggest that family scheduling constraints, particularly mid-week conflicts and early morning logistics challenges, significantly impact attendance.

3) H3: Environmental Factors

Environmental factors exhibited strong associations with attendance. Typhoon signal level demonstrated a pronounced effect: appointments during Signal 0 (no typhoon) had a 38.9% no-show rate, while those during Signal 1-2 showed 67.3% no-show rates, and Signal 3+ reached 89.2% ($\chi^2 = 127.45$, $p < 0.001$). This represents a more than two-fold increase in no-show probability during moderate typhoon conditions. Holiday status also significantly affected attendance: appointments on declared holidays had a 58.3% no-show rate compared to 40.1% on regular days ($\chi^2 = 22.34$, $p < 0.001$). Rainy season designation (June-November) showed a modest but significant effect, with 44.8% no-show rate versus 39.2% during dry months ($\chi^2 = 4.87$, $p = 0.027$).

4) H4: Demographic and Clinical Risk Profiles

Demographic variables showed mixed results. Age groups exhibited modest descriptive differences (4-6 years: 40.2%, 7-9 years: 46.1%, 10-12 years: 43.8%), but these did not reach statistical significance after formal testing ($\chi^2 = 5.12$, $p = 0.16$). Diagnosis categories similarly showed descriptive variation (ASD: 41.2%, GDD: 48.7%, SLD: 39.4%, ODD: 52.3%), but chi-square test indicated no significant difference overall ($\chi^2 = 8.73$, $p = 0.12$). Distance categories (Near <5km: 39.1%, Moderate 5-15km: 42.8%, Far >15km: 45.6%) also failed to reach significance ($\chi^2 = 3.21$, $p = 0.20$). The lack of statistical significance for these demographic factors suggests that while individual patient characteristics may contribute to risk profiles, they are less influential than environmental and temporal factors.

5) H5: Historical Behavior

Historical attendance behavior emerged as a powerful predictor. Patients with no previous no-shows had a 28.3% current no-show rate, while those with 1-2 previous no-shows showed 54.7%, and patients with 3+ previous no-shows exhibited a 78.9% current no-show rate ($\chi^2 = 156.32$, $p < 0.001$). The correlation between previous no-show count and current no-show status was strong ($r = 0.52$, $p < 0.001$). No-show ratio (proportion of previous appointments missed) similarly predicted current behavior: patients with 0% historical no-show ratio had 25.1% current no-show rate, while those with 50%+ historical ratio showed 81.4% current no-show rate. These findings strongly support the hypothesis that attendance behavior exhibits temporal consistency.

C. Predictive Model Performance

Table I presents comprehensive performance metrics for the three implemented models across the test set.

TABLE I

Model Performance Metrics on Test Set

Model	AUC-ROC	Precision	Recall	F1-Score
Logistic Regression	0.877	0.79	0.81	0.80
Random Forest	0.892	0.82	0.84	0.83
Gradient Boosting	**0.919**	**0.86**	**0.87**	**0.87**

Gradient boosting achieved the best performance with AUC-ROC = 0.919, indicating excellent discriminative ability. All three models demonstrated strong performance (AUC > 0.85), substantially exceeding baseline random classification (AUC = 0.50). The gradient boosting model's superior precision (0.86) and recall (0.87) indicate balanced performance in identifying both no-shows and attended appointments, minimizing both false positives and false negatives. ROC curves for all three models showed similar trajectories with gradient boosting maintaining slight advantages across all probability thresholds. Calibration plots revealed good agreement between predicted probabilities and observed frequencies, with minimal systematic over- or under-prediction.

D. Feature Importance Analysis

SHAP analysis on the gradient boosting model quantified the relative importance and directional effects of all predictor variables. Table II presents the top 10 features ranked by mean absolute SHAP value.

TABLE II

Top 10 Features by SHAP Importance

Rank	Feature	Mean |SHAP|
1	Typhoon Signal Level	0.182
2	No-Show Ratio	0.156
3	Log Previous Appointment Count	0.134
4	Log Lead Time Days	0.118
5	Age	0.095
6	Is Weekend	0.087
7	Day of Week (Wednesday)	0.078
8	Is Holiday	0.072
9	Time of Day (Morning)	0.065

10	Log Distance KM	0.058

Typhoon signal level emerged as the single most influential predictor (mean |SHAP| = 0.182), confirming the dominant role of weather-related barriers in attendance decisions. Higher signal levels consistently increased no-show probability, with the effect becoming particularly pronounced at Signal 2 and above. Historical behavior variables (no-show ratio, previous appointment count) ranked second and third, reinforcing the importance of past patterns. Notably, financial variables (down payment status, payment amount) did not appear in the top 10 features, consistent with the hypothesis testing results showing no significant effect. Weekend status and specific day (Wednesday) demonstrated meaningful but moderate effects, while distance showed relatively weak influence, suggesting that geographic proximity is less critical than other factors once patients have established a relationship with the clinic.

E. High-Risk Profile Identification

Analysis of predictions revealed specific appointment profiles associated with elevated risk. The highest-risk scenarios combined multiple adverse factors: weekday (especially Wednesday) appointments, morning time slots, typhoon warnings (Signal 2+), patients with poor historical attendance (noshow_ratio >50%), and longer lead times (>14 days). For instance, a Wednesday morning appointment during Signal 2 conditions for a patient with 60% historical no-show rate yielded a predicted 86.3% no-show probability using logistic regression and 97.5% using gradient boosting. Conversely, the lowest-risk profiles featured weekend appointments, afternoon timing, normal weather conditions, patients with perfect historical attendance, and short lead times (<3 days). These profiles exhibited predicted no-show probabilities below 20%. The gradient boosting model demonstrated particular precision in identifying extreme risk cases, with predicted probabilities above 90% showing actual no-show rates of 94.7%, while predicted probabilities below 20% showed actual no-show rates of only 12.3%.

V. DISCUSSION

A. Interpretation of Key Findings

The study's findings reveal a complex attendance behavior landscape dominated by environmental and historical factors rather than financial incentives. The emergence of typhoon signal level as the top predictor underscores the profound impact of tropical weather systems on healthcare access in the Philippines. The dramatic increase in no-show rates from 38.9% under normal conditions to 89.2% during Signal 3+ warnings demonstrates that severe weather creates insurmountable barriers regardless of patient commitment or clinic policies. This finding has important implications: it suggests that weather-related cancellations may be largely unavoidable and should be accommodated through flexible rescheduling policies rather than penalized through strict cancellation fees.

The strong predictive power of historical attendance behavior validates theories of behavioral consistency but also raises important questions about intervention timing. Patients with established no-show patterns (>50% historical rate) showed current no-show rates exceeding 80%, suggesting that once this pattern develops, it becomes self-reinforcing. This may reflect accumulated frustration with scheduling difficulties, loss of treatment motivation, or systematic life circumstances (such as unstable employment or transportation) that persist over time. The finding implies that interventions targeting at-risk patients must occur early in the treatment relationship, before negative patterns solidify.

Perhaps the most surprising finding is the complete absence of financial commitment effects. The lack of significant difference between appointments with and without down payments contradicts economic incentive theory and challenges conventional wisdom in healthcare management. Several explanations warrant consideration. First, the down payment amounts (₱150-₱1,500) may be too small relative to the full session cost (typically ₱2,000-₱3,000) to create meaningful sunk cost psychology. Second, families facing genuine barriers to attendance (weather, illness, transportation failure) may miss appointments regardless of financial loss, suggesting that non-attendance is often involuntary rather than a discretionary choice. Third, the payment structure may not create psychological commitment if families view deposits as merely shifting payment timing rather than adding cost. This finding suggests that increasing deposit requirements, a common clinic policy response to no-shows, may be ineffective and could create access barriers without improving attendance.

The temporal patterns revealing higher Wednesday and morning no-show rates likely reflect the intersection of work, school, and family schedules. Wednesdays fall mid-week when work and school obligations are fully engaged, unlike Mondays when schedules are still flexible or Fridays when early dismissals are more common. Morning appointments may conflict with school drop-off logistics, commute timing from residential areas, or parental work start times. The lower weekend no-show rates suggest that when temporal flexibility exists, families prioritize therapy attendance, indicating genuine commitment despite high overall no-show rates. This finding supports scheduling policies that preferentially allocate weekend and afternoon slots to patients with higher baseline risk factors.

B. Comparison to Prior Research

The current study's model performance (AUC = 0.919 for gradient boosting) exceeds the 0.75-0.80 range reported by Dunstan et al. [1] in the Chilean pediatric setting, despite using a smaller dataset. This superior performance likely reflects the inclusion of highly predictive environmental variables absent from the Chilean study. The confirmation that historical attendance behavior strongly predicts future outcomes aligns with Dunstan's findings, suggesting this relationship is robust across cultural contexts. However, the current study's identification of typhoon risk as the dominant predictor introduces a factor entirely missing from prior research, highlighting the importance of context-specific variables.

The temporal patterns observed partially align with Hathaway et al.'s [2] findings regarding time-of-day effects, though the specific pattern differs. While Hathaway reported higher no-shows in both early morning and late afternoon, the current study found morning risk concentrated in earlier hours with afternoon appointments showing better attendance. This difference may reflect cultural variations in family scheduling or may stem from the Philippine practice of extended family support for childcare, which becomes more available in afternoon hours. The methodological use of SHAP analysis for interpretability mirrors Hathaway's approach and demonstrates its value in clinical contexts where model transparency is essential for user acceptance.

The environmental factor findings extend Liu et al.'s [3] work on weather variables but with a crucial difference in magnitude. While Liu found that severe weather increased no-show probability by up to 40%, the current study observed increases exceeding 100% (from 38.9% to 89.2%), more than doubling the baseline rate. This dramatic difference likely reflects the distinct nature of Philippine typhoons compared to U.S. winter storms. Typhoons create compound barriers including flooding, power outages, and government-mandated movement restrictions that rarely occur simultaneously in temperate weather events. This finding emphasizes that climate-specific variables require localized calibration rather than assumption of universal effects.

The absence of financial effects contrasts sharply with assumptions underlying Mochón et al.'s [4] work, which presumed that economic factors would moderate historical behavior patterns. The current findings suggest that in contexts where out-of-pocket payment is standard, financial mechanisms may operate differently than in insurance-based systems. This challenges the generalizability of Western healthcare management strategies and implies that interventions effective in one payment context may fail in another.

C. Practical Implications for Clinic Management

The research findings generate several actionable recommendations for clinic operations. First, the gradient boosting model's strong performance (AUC = 0.919) and precise high-risk identification support deployment as a real-time decision support tool. The clinic could implement automated risk scoring for all scheduled appointments, flagging those exceeding 60% predicted no-show probability for targeted intervention. Such interventions might include confirmation calls 48 hours prior, SMS reminders, or offers to reschedule to lower-risk time slots. The model's ability to identify appointments with >90% predicted no-show probability with 94.7% accuracy enables proactive overbooking of these slots, filling therapist schedules more efficiently.

Second, the dominance of environmental factors suggests that weather-responsive policies would yield greater attendance improvements than financial penalties. The clinic should implement flexible rescheduling during typhoon warnings (Signal 2+) without cancellation fees, recognizing that these conditions create force majeure circumstances. Automatic rescheduling notifications could be triggered by PAGASA weather bulletins, reducing administrative burden while demonstrating patient-centered care. Similarly, holiday periods might warrant reduced scheduling or enhanced reminder protocols.

Third, the temporal pattern findings support strategic schedule design. Given the lower weekend no-show rates (33.4% vs. 45.8% weekdays), the clinic could expand Saturday operations and reduce Wednesday scheduling. High-priority appointments requiring guaranteed attendance (e.g., initial evaluations, specialized assessments) should be preferentially scheduled for weekends and afternoons. Conversely, routine follow-up sessions might be concentrated in higher-risk weekday morning slots where some no-shows can be absorbed without disrupting clinic flow. This strategic allocation would maximize utilization of reliable time blocks.

Fourth, the strong influence of historical behavior suggests that early intervention with at-risk patients could prevent establishment of chronic no-show patterns. Patients missing their second or third appointment (before patterns solidify) should receive personal outreach from clinic coordinators to identify and address barriers. This might reveal systematic obstacles (transportation issues, work schedule conflicts, financial constraints) that can be resolved through solutions such as modified scheduling, payment plans, or telehealth options for follow-ups. Early intervention investment could prevent the development of high-risk profiles that prove extremely difficult to remediate later.

Fifth, the absence of financial effects suggests that current deposit policies should be reevaluated. Rather than increasing deposit requirements (which may create access barriers without improving attendance), the clinic might experiment with alternative commitment mechanisms such as signed attendance agreements, treatment milestone tracking, or positive reinforcement for consistent attendance (e.g., fee discounts for patients maintaining >90% attendance rates). These approaches might generate psychological commitment without financial exclusion.

D. Study Limitations

Several methodological limitations warrant acknowledgment. First, the single-clinic sample restricts generalizability to other Philippine pediatric therapy settings. The observed patterns may reflect idiosyncrasies of this clinic's operations, patient population, or local conditions. The clinic's specific location within Metro Manila, its payment structure, and its patient demographics may not represent other facilities. Multi-clinic validation across diverse geographic regions, urban-rural settings, and clinic types would be necessary to establish broader applicability.

Second, while the dataset provides sufficient observations for model development (n=1,506), the sample size limits statistical power for detecting modest effects, particularly in subgroup analyses. The finding that demographic variables (age groups, diagnosis categories) showed no significant effects may reflect insufficient power to detect real but small differences rather than true absence of effects. Larger datasets would enable more definitive conclusions about whether certain patient populations exhibit systematically different attendance patterns.

Third, unmeasured confounding variables may influence observed associations. For example, family stress levels, caregiver mental health, transportation mode, and childcare availability likely affect attendance but remain uncaptured in administrative data. The observed relationship between historical no-shows and current attendance may partially reflect persistent underlying stressors rather than purely behavioral patterns. Similarly, the lack of financial effects might be confounded by unmeasured economic factors; families experiencing severe hardship might both struggle to make deposits and face multiple attendance barriers, obscuring the deposit's independent effect.

Fourth, measurement error affects several variables. Residential distance calculations based on declared addresses may not reflect actual commute origins if parents travel from workplaces. Diagnosis categories derived from intake forms may lack the precision of formal clinical assessments. Down payment recording may have inconsistencies if some payments were recorded with delays or misclassified. These measurement issues could introduce noise that attenuates true relationships or creates spurious associations.

Fifth, the observational design precludes definitive causal inference despite statistical associations. The model identifies predictive patterns but cannot prove that intervening on predictor variables would change outcomes. For instance, while down payments show no association with attendance, this does not definitively establish that implementing mandatory deposits would be ineffective; selection effects might obscure causal relationships if families who voluntarily make deposits differ systematically from those who do not in ways beyond what measured variables capture.

Sixth, temporal dynamics may limit model stability. The data collection period's specific weather patterns, holiday calendar, and local circumstances may not represent longer-term averages. Economic conditions, policy changes, or clinic operational modifications could alter attendance patterns over time, requiring periodic model retraining. The cross-sectional nature of the data prevents assessment of whether predictive relationships remain stable or evolve.

E. Recommendations for Future Research

Future research should address current limitations through expanded scope and refined methodology. Multi-clinic studies encompassing diverse Philippine regions, clinic types, and patient populations would test generalizability and identify context-dependent versus universal predictors. Particular value would come from comparing private clinics with different payment structures, public facilities, and insurance-based systems to isolate the effects of financial mechanisms.

Longitudinal designs tracking patients over extended periods would enable examination of attendance trajectory dynamics. Such studies could test whether patients exhibit stable profiles or whether behavior evolves in response to treatment progress, life circumstances, or clinic interventions. Dynamic modeling approaches could assess whether risk factors interact with treatment duration, revealing if predictive patterns differ for new versus established patients.

Incorporating additional predictor variables would enrich explanatory power. Psychosocial factors (parental stress, family structure, caregiver mental health, social support) could be captured through validated survey instruments. Transportation mode and commute time might prove more informative than residential distance. Treatment engagement measures (homework completion, goal progress) could indicate commitment beyond attendance alone. Socioeconomic variables (household income, employment status, education level) would enable more sophisticated analysis of financial effects.

Experimental or quasi-experimental designs would establish causal effects where observational methods cannot. Randomized controlled trials of intervention strategies informed by the prediction model could test whether targeted reminders, flexible scheduling, or modified payment policies actually reduce no-shows. Natural experiments exploiting policy changes would provide causal estimates using difference-in-differences or regression discontinuity designs. For instance, clinics altering deposit requirements could serve as intervention sites compared to matched controls.

Qualitative research would complement quantitative modeling by revealing decision-making processes underlying attendance behavior. In-depth interviews with families who frequently cancel could identify barriers and motivations not captured by structured data. Focus groups with clinic staff could uncover operational challenges in implementing prediction-based management. Mixed-methods approaches combining model predictions with qualitative case studies would provide rich understanding of how statistical patterns manifest in individual families' experiences.

Extension to other healthcare contexts would test model transferability. Application to general pediatric care, adult therapy services, or chronic disease management could reveal domain-specific versus universal dynamics. Cross-national comparisons with other Southeast Asian countries sharing similar climate and cultural characteristics would illuminate region-specific versus global predictors.

Finally, cost-effectiveness analysis would justify investment in prediction systems. Comparing implementation costs (model development, software integration, staff training) against benefits (reduced therapist idle time, increased revenue, improved patient outcomes) would determine return on investment. Economic evaluation would support evidence-based decisions about technology adoption in resource-constrained settings.

VI. CONCLUSION

This research successfully developed a high-performing machine learning model (AUC = 0.919) for predicting appointment no-shows in a Philippine pediatric therapy clinic, transforming routine administrative data into actionable intelligence for clinic management. The study's key contribution lies in demonstrating that contextually relevant environmental variables, particularly typhoon warnings, dominate attendance predictions in tropical settings, fundamentally differentiating Philippine healthcare dynamics from patterns documented in temperate Western systems. The finding that typhoon signal level emerged as the single most influential predictor, increasing no-show rates from 39% to 89% during severe conditions, underscores the profound impact of climate-specific factors on healthcare access.

The research challenges conventional assumptions about financial incentives in healthcare adherence. The complete absence of significant effects from down payment requirements contradicts economic incentive theory and questions the efficacy of deposit-based policies commonly implemented by clinics. This finding suggests that attendance barriers in the Philippine private therapy context are predominantly external and involuntary (weather, scheduling conflicts) rather than reflecting discretionary choices responsive to financial stakes. The implication is that management strategies emphasizing flexible accommodation of genuine barriers may prove more effective than policies focused on financial penalties or commitments.

The strong predictive power of historical attendance behavior, while validating behavioral consistency theories, simultaneously reveals the urgency of early intervention with at-risk patients. The finding that patients with established poor attendance patterns (>50% historical no-show rate) show current no-show rates exceeding 80% indicates that once negative patterns develop, they become self-reinforcing and difficult to remediate. This suggests that clinic resources should prioritize identifying and addressing barriers during patients' initial appointments, before chronic patterns solidify.

The practical applications of these findings are immediate and concrete. The gradient boosting model's ability to identify appointments with >90% predicted no-show probability with 95% accuracy enables sophisticated schedule management including strategic overbooking, targeted reminder protocols, and preferential allocation of reliable time slots (weekends, afternoons) to high-priority appointments. The identification of specific high-risk profiles combining mid-week timing, morning slots, typhoon warnings, and poor historical attendance provides clear targets for intervention resource allocation.

Beyond operational benefits, the study contributes methodologically by demonstrating the value of explainable AI techniques (SHAP analysis) in clinical contexts. The ability to quantify not merely which factors predict no-shows but precisely how much influence each factor exerts and in what direction facilitates user acceptance and informed decision-making. Clinic administrators can understand and trust model recommendations because the reasoning is transparent rather than opaque.

The research acknowledges important limitations including single-clinic sampling, potential unmeasured confounding, and inability to establish definitive causal relationships from observational data. However, these limitations delineate boundaries of inference rather than invalidate contributions. They point toward productive directions for future research including multi-clinic validation, experimental intervention trials, incorporation of psychosocial variables, and longitudinal tracking of attendance trajectories.

The broader significance extends beyond pediatric therapy to healthcare delivery in tropical developing countries more generally. The dominance of weather-related barriers and the relative unimportance of financial mechanisms may characterize healthcare access patterns throughout Southeast Asia and similar regions, suggesting that management strategies developed in temperate Western contexts require substantial adaptation rather than direct transfer. The study demonstrates that effective evidence-based management demands locally grounded research that captures context-specific realities.

Ultimately, this project exemplifies how systematic analysis of routine administrative data using appropriate machine learning techniques can transform operational challenges into opportunities for evidence-based improvement. The prediction model and associated insights provide Philippine pediatric therapy clinics with both a functional tool for day-to-day schedule management and deeper understanding of the factors shaping patient behavior in this setting. As healthcare systems worldwide face mounting pressure to improve efficiency amid growing demand, such data-driven approaches to operational optimization become not merely beneficial but essential for sustainable delivery of quality care.

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